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Tube Calibration

Mantid Tools for calibrating tubes

Karl Palmen and Gesner Passos

Adapted from work proposed by Rob Bewley and Toby Perring

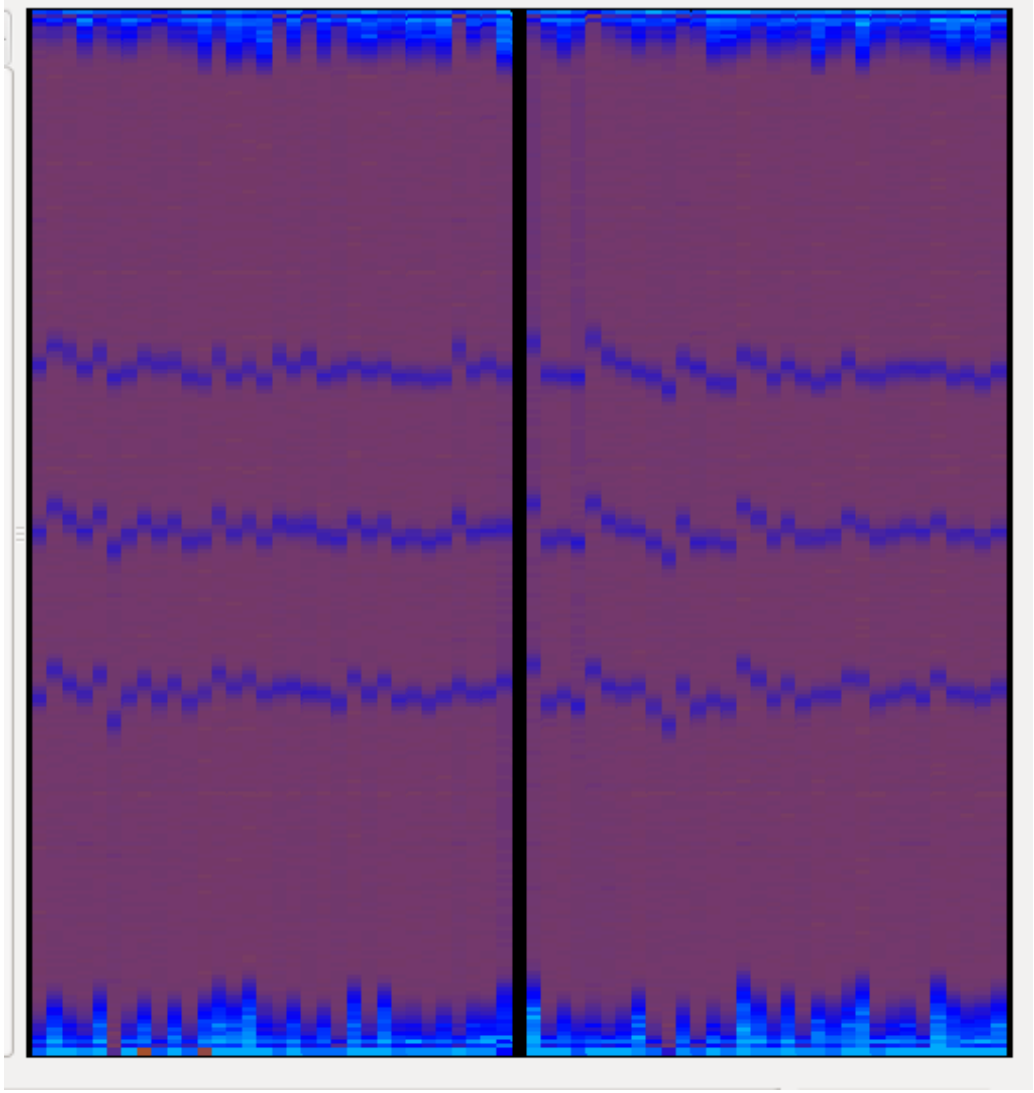
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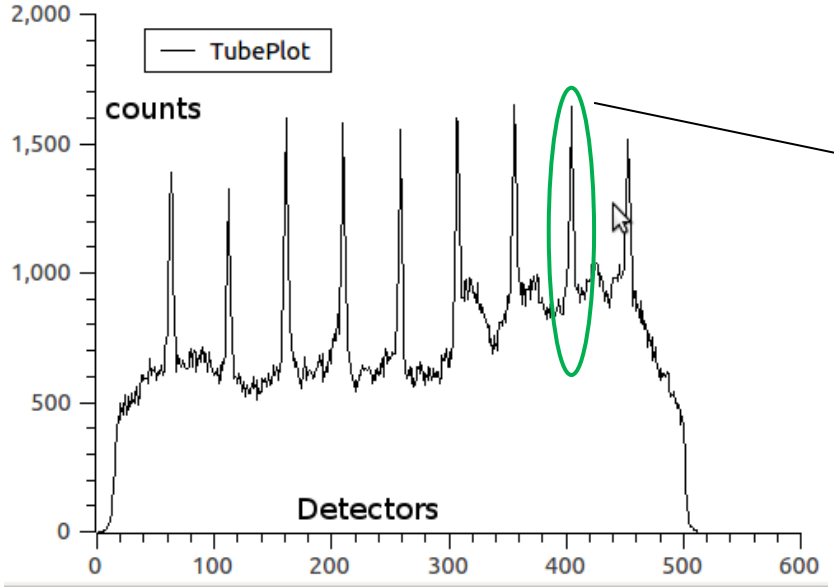
Introduction

Uncalibrated Tubes

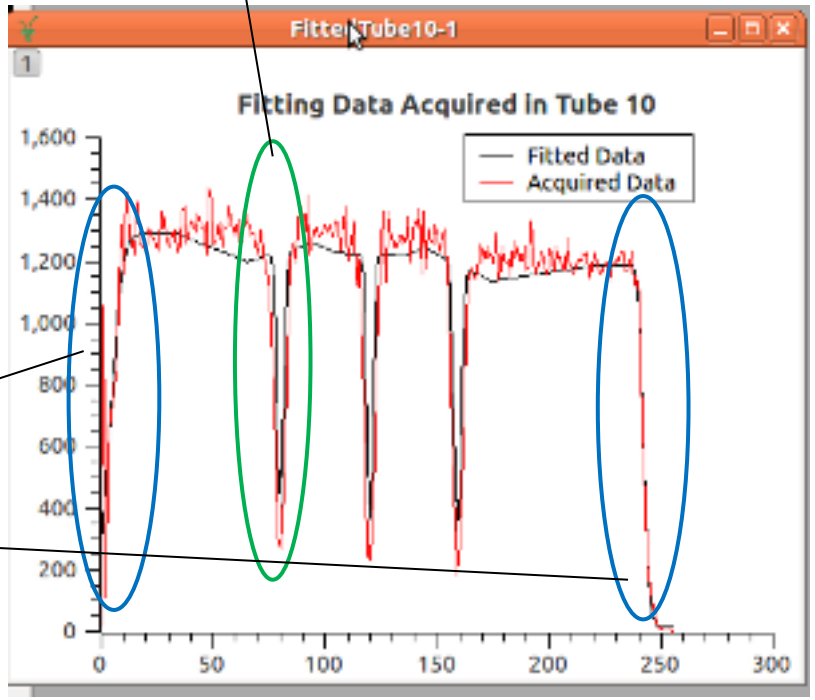


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Techinique – Finding Peak Position



Gaussian



Edges

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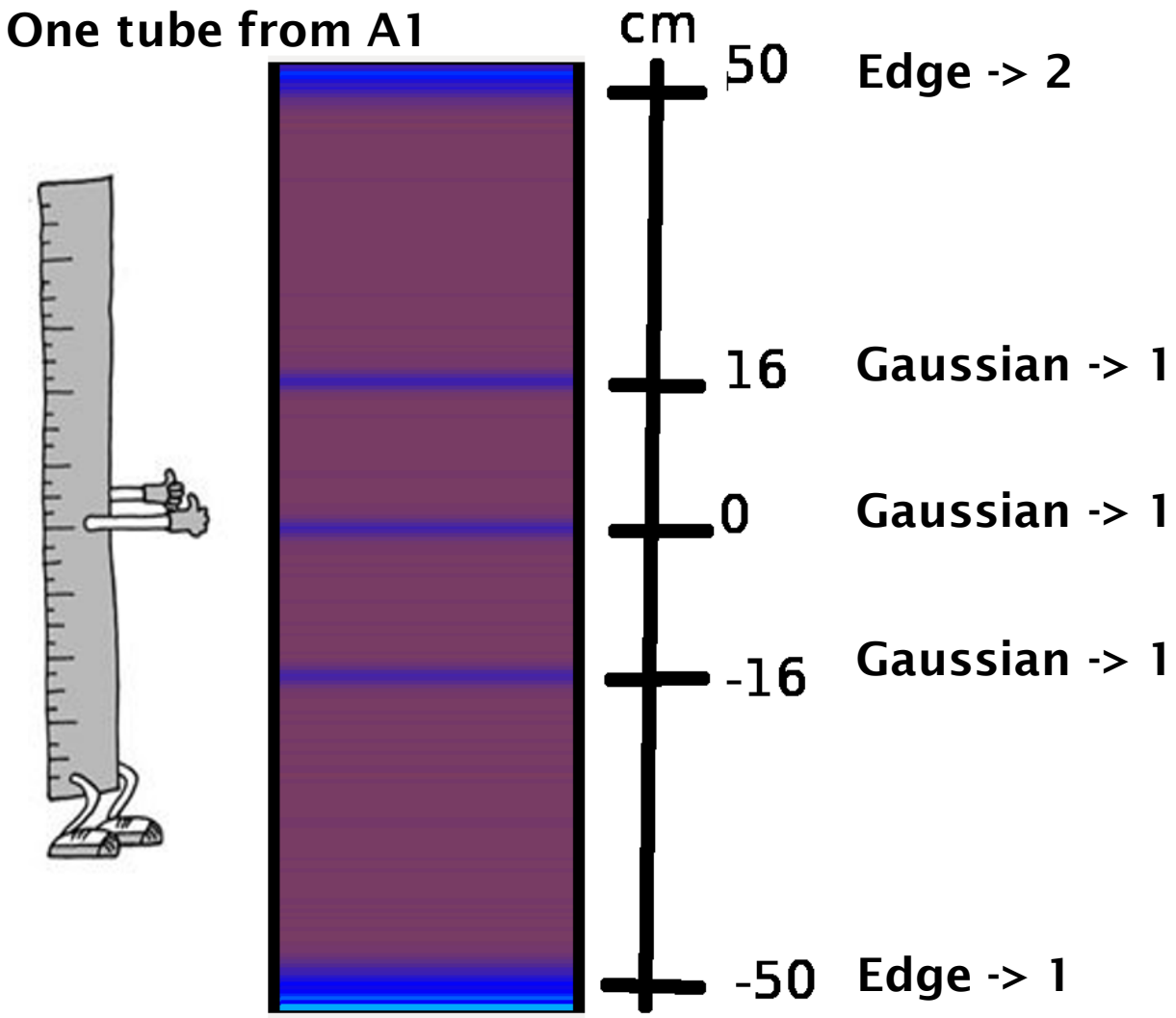
Python Code for the calibration

```
calibrate(workspace,
          tubeSet,          # define the group of tubes
          knownPositions, # position of the peaks
          funcForm           # shape of the peaks

[, fitPar, margin,        # optionals
rangeList, calibTable,
plotTube, excludeShorTubes,
overridePeaks, fitPolyn,

outputPeak])
```

Example 1- MAPS door A1



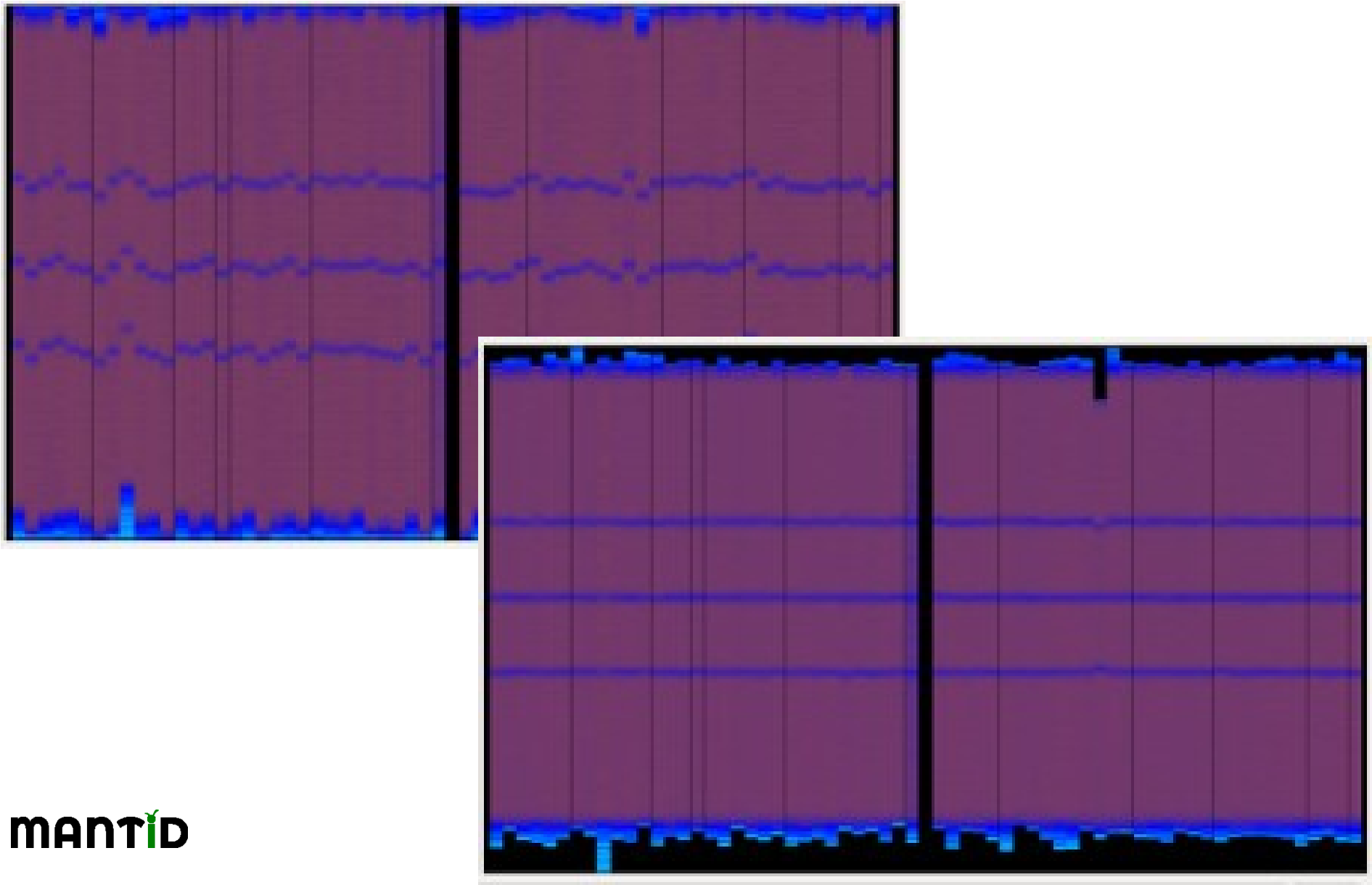
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Example 1- MAPS door A1

```
> from tube import calibrate
> ws = Load('MAP14919')
> ws = Integration(ws, RangeLower=2000,
                       RangeUpper=10000)
> known_pos = [-0.50, -0.16, 0.0, 0.16, 0.50]
> peaks_form = [2, 1, 1, 1, 2]
> calibTable = calibrate(ws,
     'MAPS/A1_window', known_pos,
                       peaks_form)
>
> ApplyCalibration(ws, calibTable)
```

MAPS door A1 calibrated



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Options (1)

- **fitPar** – Define the parameters to be used in the fit. If not provided, the dynamic mode is used.
- **margin** –value in pixesl that will be used around the peaks/edges to fit them. Default = 10

```
>>> fit_start, fit_end = centre - margin, centre + margin
```

- **rangeList** –list of tubes indexes that will be calibrated. As in the following code:

```
>>> for index in rangelist:
>>> do_calibrate(tubeSet.getTube(index))
```

http://download.mantidproject.org/docs/master/python/html/techiniques/calibration.html

Options (2)

- **calibTable** – Pass the calibration table, it will them append the values to the provided one and return it.
- **plotTube** – If given, the tube whose index is in plotTube will be ploted as well as its fitted peaks. (It can receive a list of indexes to plot)
- **excludeShortTubes** – Do not calibrate tubes whose length is smaller than given value.
- **overridePeaks** –dictionary that defines an array of peaks positions (in pixels) to be used for the specific tube(key).
- **fitPolyn** – Order of the polynomial to distribute the detectors.

Options (3)

- **outputPeak** –Enable the calibrate to output the peak table, relating the tubes with the pixels positions. It may be passed as a boolean value (outputPeak=True) or as a peakTable value. The later case is to inform calibrate to append the new values to the given peakTable. This is usefull when you have to operate in subsets of tubes.
- **>>>** calibTable, peakTable = calibrate(ws, (omitted), rangeList=[1], outputPeak=True)
- **>>>** calibTable, peakTable = calibrate(ws, (omitted), rangeList=[2], outputPeak=peakTable)

http://download.mantidproject.org/docs/master/python/html/techiniques/calibration.html



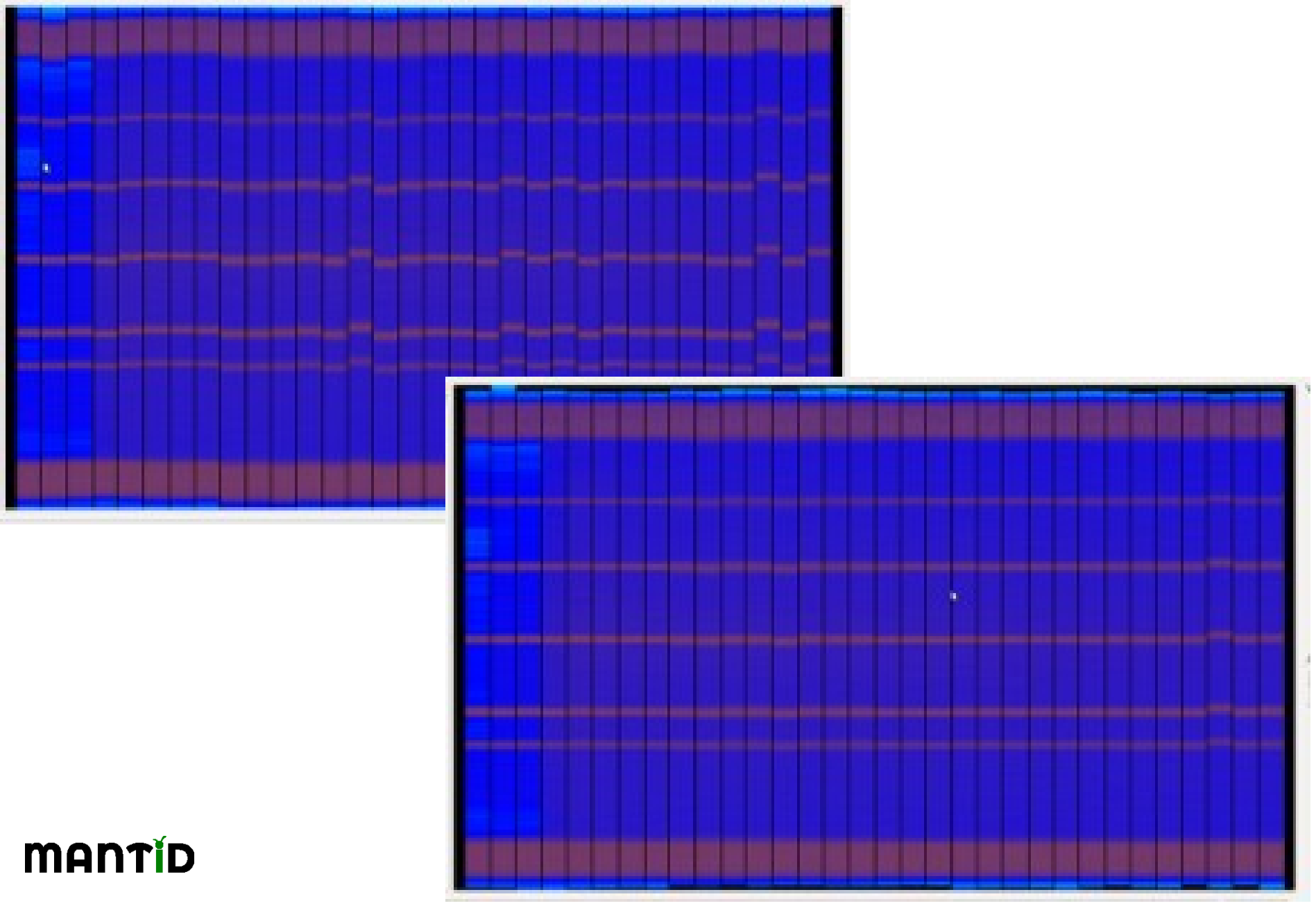
Example 2- MERLIN door1 and door2

```
> from tube import calibrate
> ws = Load('MER12024')
> ws = Integration(ws, RangeLower=3000,
                       RangeUpper=20000)
> known_pos = [-1.39, -1.20, -0.62, -0.43,
               0.00 , 0.43, 0.82, 1.20, 1.39]
> peaks_form = [2, 2, 1, 1, 1, 1, 1, 2, 2]

> calibTable = calibrate(ws, [‘door1‘, ‘door2‘],
                         known_pos, peaks_form)

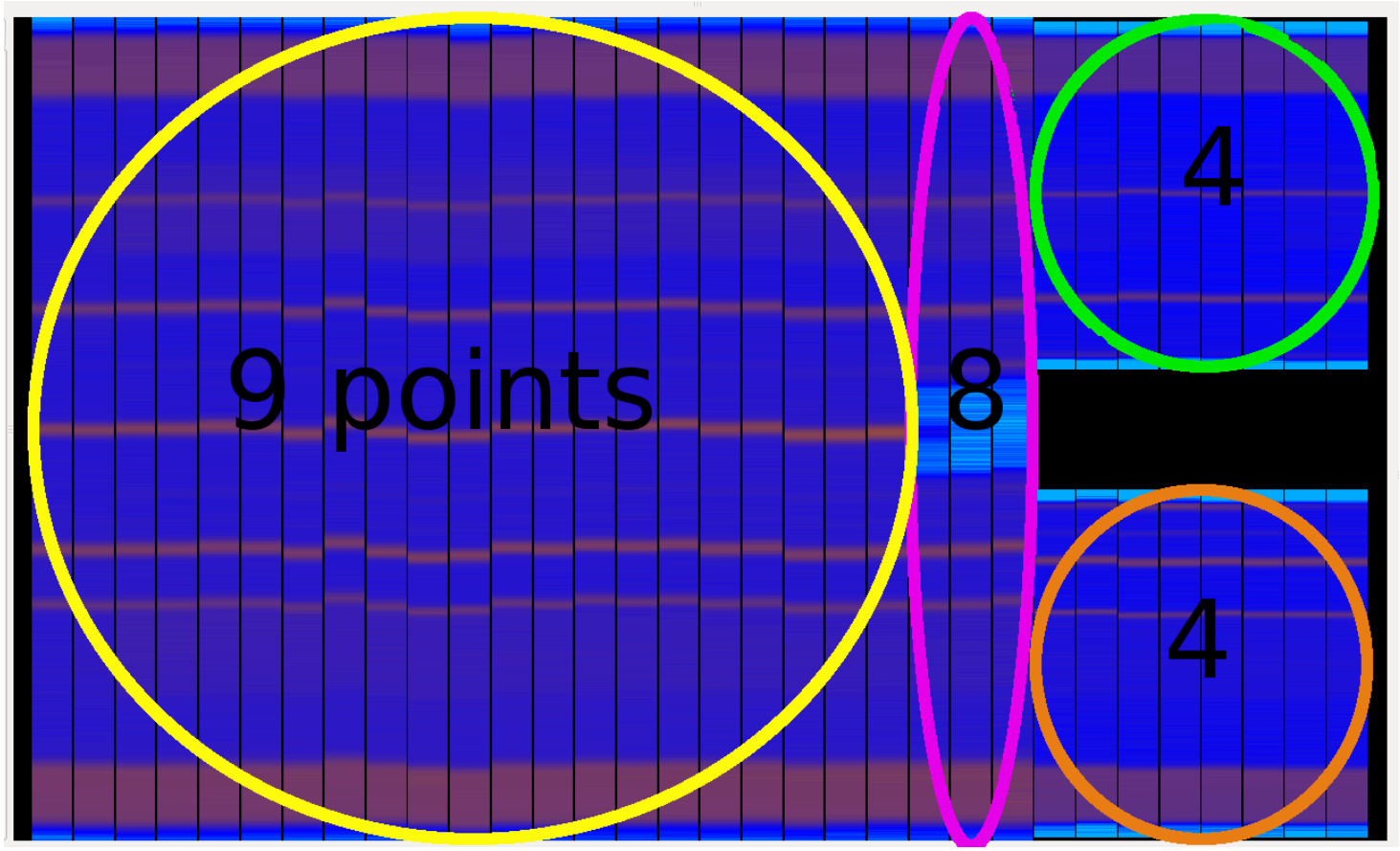
> ApplyCalibration(ws, calibTable)
```

MERLIN door 1 calibrated



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MERLIN door3

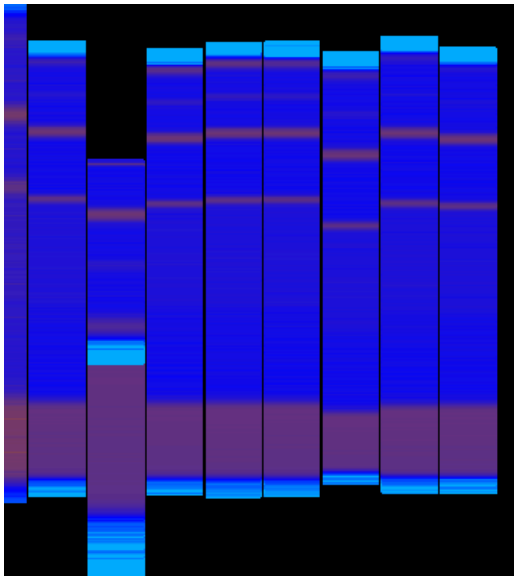


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Example 3 – MERLIN door 3 small tubes

```
> from tube import calibrate
> ws = Load('MER12024')
> ws = Integration(ws, RangeLower=3000,
                       RangeUpper=20000)
> tubes = ['MERLIN/door3/tube_1_%d'%(i) for i in
                                    range(1,9)]
> known_pos = [-0.55, -0.36, 0.21, 0.41]
> peaks_form = [2, 2, 1, 1]

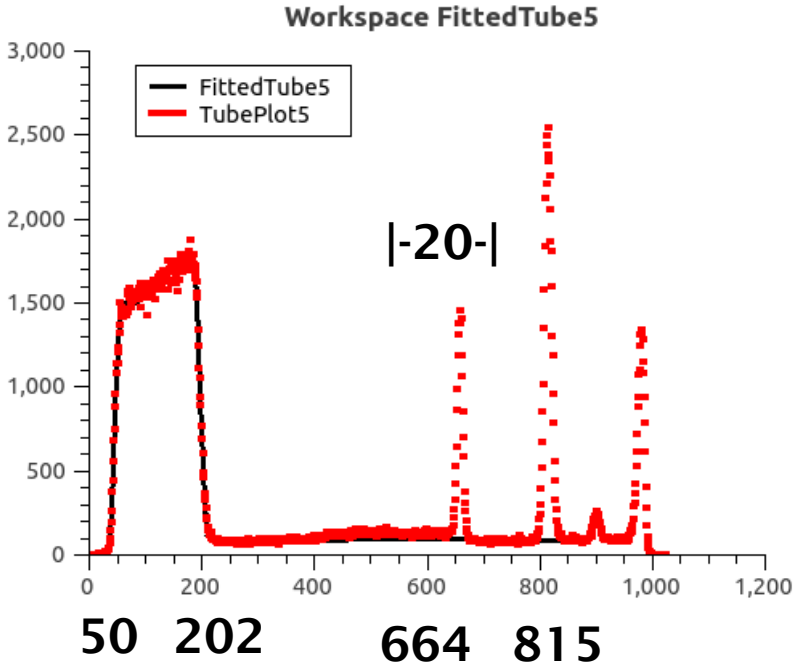
> calibTable = calibrate(ws, tubes,
                known_pos, peaks_form)
> ApplyCalibration(ws, calibTable)
```



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Example 3 – MERLIN door 3 small tubes

```
> from tube import calibrate
> ws = Load('MER12024')
> ws = Integration(ws, RangeLower=3000,
>                      RangeUpper=20000)
> known_pos = [-0.55, -0.36, 0.21, 0.41]
> peaks_form = [2, 2, 1, 1]
> tubes = ['MERLIN/door3/tube_2_%d'%(i) for i in range(1,9)]
> calibTable = calibrate(ws, tubes,
                         known_pos, peaks_form, plotTube=[1,3,5])
> ApplyCalibration(ws, calibTable)
```

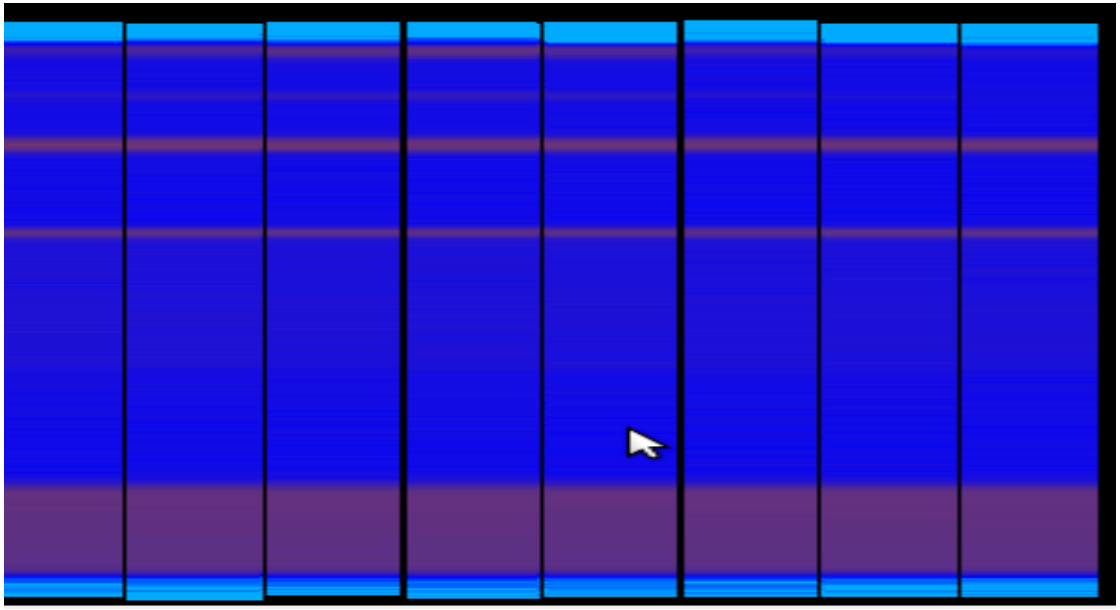


Auto Finding Peaks does not work for this example!
Let's specify where the peaks are.

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Example 3 – MERLIN door 3 small tubes

```
> from tube import calibrate
> from tube_calib_fit_params import TubeCalibFitParams
> ws = Load('MER12024')
> ws = Integration(ws, RangeLower=3000,
>                      RangeUpper=20000)
> known_pos = [-0.55, -0.36, 0.21, 0.41]
> peaks_form = [2, 2, 1, 1]
> tubes = ['MERLIN/door3/tube_1_%d'%(i) for i in range(1,9)]
> fitParam = TubeCalibFitParams([50, 202, 664, 815] )
> calibTable = calibrate(ws, tubes, known_pos, peaks_form,
>                        fitPar=fitParam, margin=20 )
> ApplyCalibration(ws, calibTable)
```



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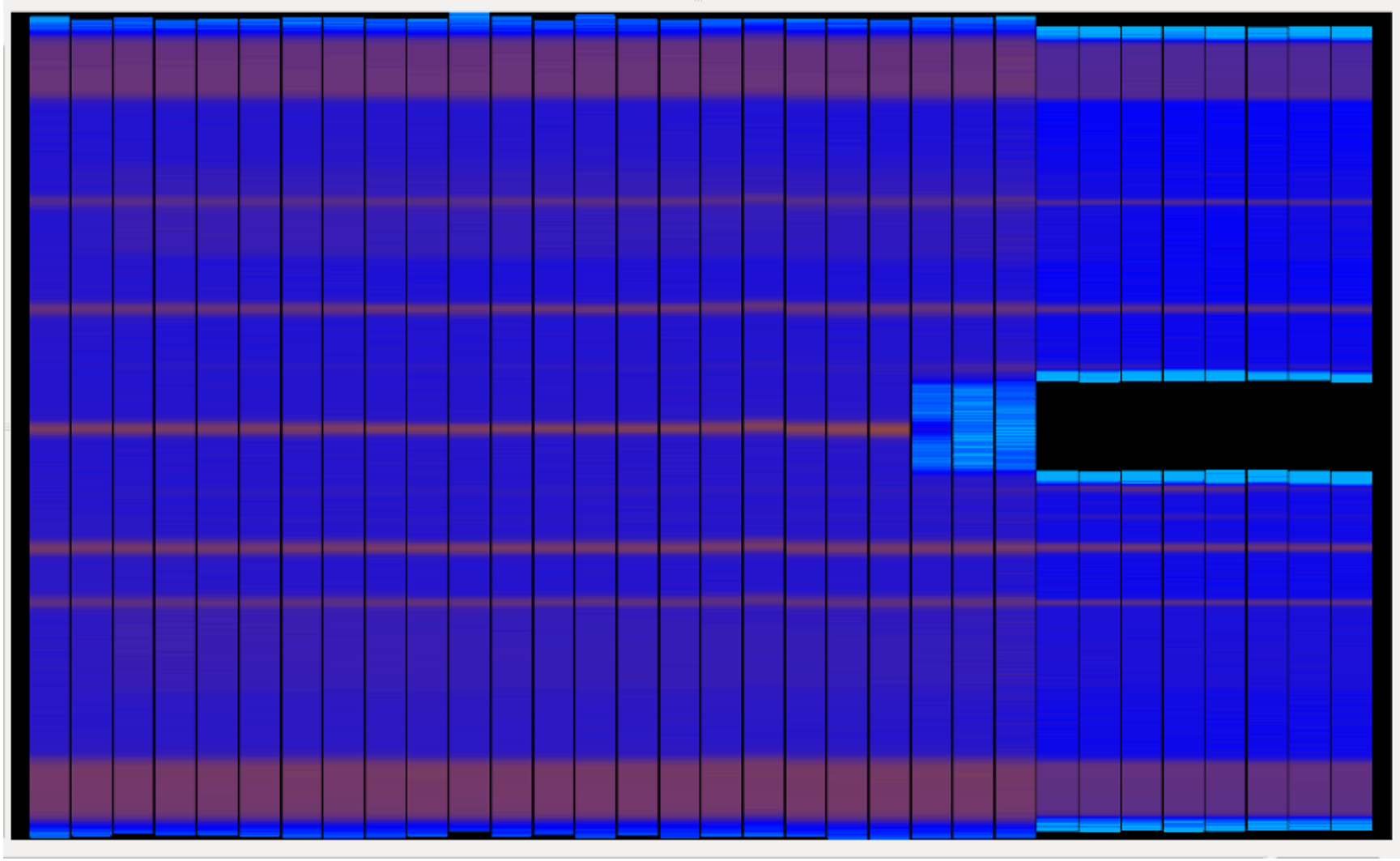
Example 4 – MERLIN door 3

Don't Read the code!

```
from tube import calibrate
from tube_calib_fit_params import TubeCalibFitParams
ws = LoadRaw('MER12024.raw')
ws = Integration(ws, RangeLower=3000, RangeUpper=20000)
# calibrate 4 points down
kn_down, ff_down = [-0.40, -0.02, 0.36, 0.55], [1,1,2,2]
fitPar_down = TubeCalibFitParams([216, 527, 826, 989], margin=20)
fitPar_down.setAutomatic(True)
small_down = ['MERLIN/door3/tube_1_%d'%(i) for i in range(1,9)]
calibrationTable = calibrate(ws, small_down, kn_down, ff_down, fitPar=fitPar_down)
# calibrate 4 points upper
kn_upper, ff_upper = [-0.55, -0.36, 0.21, 0.41] , [2, 2, 1, 1]
small_upper = ['MERLIN/door3/tube_2_%d'%(i) for i in range(1,9)]
fitPar_upper = TubeCalibFitParams([50, 202, 664, 815], margin=20)
fitPar_upper.setAutomatic(True)
calibrationTable = calibrate(ws, small_upper, kn_upper, ff_upper, fitPar=fitPar_upper,
                             calibTable=calibrationTable)
# calibrate 8 points tubes
kn8, ff8 = [-1.39, -1.20, -0.62, -0.43, 0.43, 0.82, 1.20, 1.39], [2,2,1,1,1,1,2,2]
big_8_points = ['MERLIN/door3/tube_3_%d'%(i) for i in [1,2,3]]
calibrationTable = calibrate(ws, big_8_points, kn8, ff8, calibTable=calibrationTable)
# calibrate 9 points tubes
Kn9, ff9 = [-1.39, -1.20, -0.62, -0.43, 0.00, 0.43, 0.82, 1.20, 1.39], [2,2,1,1,1,1,1,2,2]
big_9_points = ['MERLIN/door3/tube_3_%d'%(i) for i in [4,5,6,7,8]] + ['MERLIN/door3/tube_4_%d'%(i) for i in range(1,9)] + ['MERLIN/door3/tube_5_%d'%(i) for i in range(1,9)]
calibrationTable = calibrate(ws, big_9_points, kn9, ff9, calibTable=calibrationTable)
ApplyCalibration( Workspace=ws, PositionTable=calibrationTable)
```

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Example 4 - MERLIN door 3



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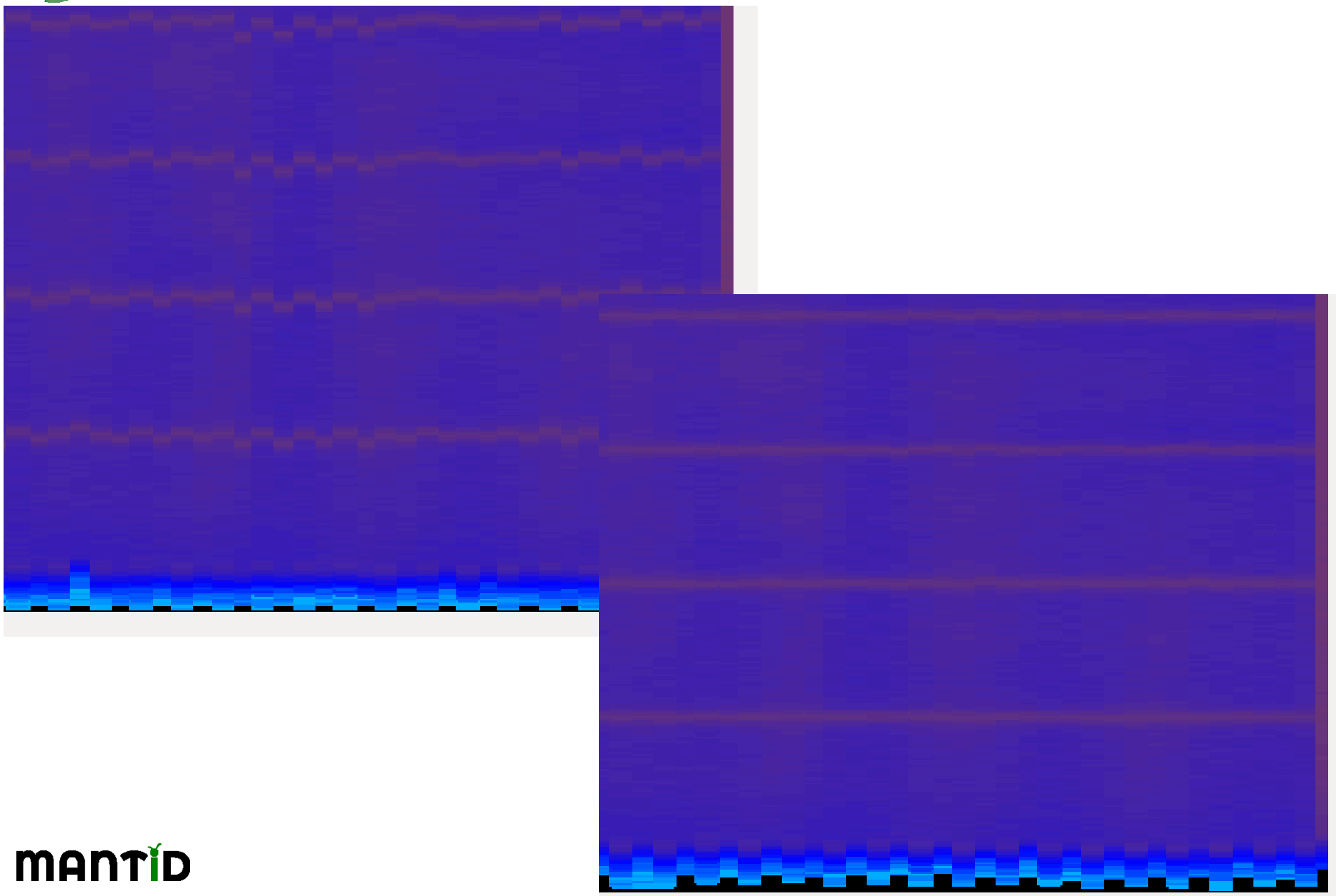


Example 5 - WISH

```
> from tube import calibrate
> import numpy
> ws = Load('WISH00017701')
> ws = Integration(ws, RangeLower=1, RangeUpper=20000)
> lower_tube = numpy.array([-0.41, -0.31, -.021, -0.11,
                            -0.02, 0.09, 0.18, 0.28, 0.39])
> upper_tube = lower_tube + 0.003
> funcForm = [1]*9
> calibrationTable = calibrate(ws, 'WISH/panel03', lower_tube,
                    funcForm, rangeList=range(0,76), margin=15)
> calibrationTable = calibrate(ws, 'WISH/panel03', upper_tube,
                    funcForm, rangeList=range(76,152), margin=15,
                            calibTable=calibrationTable)
> ApplyCalibration(ws, calibrationTable)
```

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WISH panel 3 calibrated



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Thanks

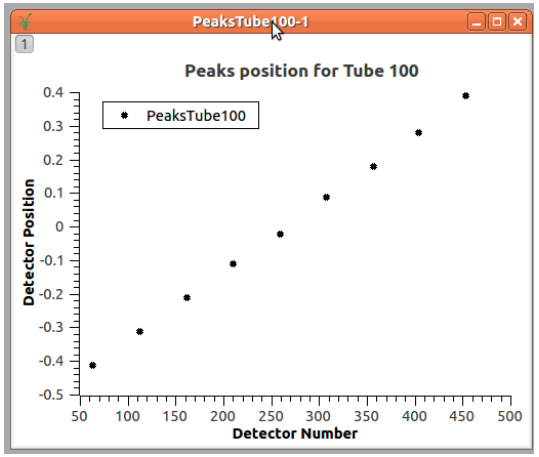
- Further documentation:

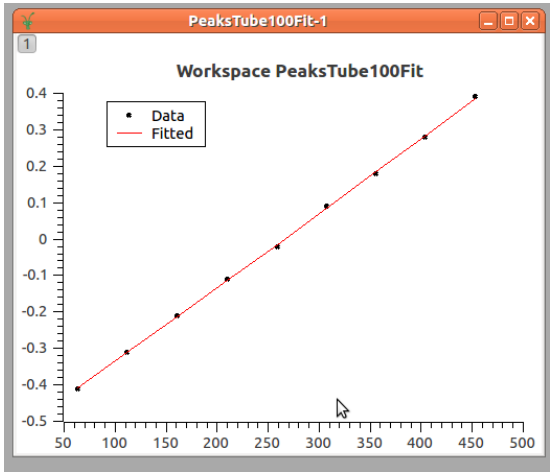
http://www.mantidproject.org/Tube_Calibration

http://download.mantidproject.org/docs/master/python/html/techiniques/calibration.html

Techinique – Positioning along the tube (2)

- Define an equation to move all key points to the defined positions
- Use this equation to all detectors
- Applies the transformation to the detectors moving it.





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